An expert knowledge based methodology for online detection of signal oscillations

Detection of valve oscillations for the CERN cryogenics control system

Filippo Tilaro
European Organisation for Nuclear Research, *CERN*
CH-1211 Geneva 23, Switzerland

Benjamin Bradu
European Organisation for Nuclear Research, *CERN*
CH-1211 Geneva 23, Switzerland

Manuel Gonzalez-Berges
European Organisation for Nuclear Research, *CERN*
CH-1211 Geneva 23, Switzerland

Mikhail Roshchin
Siemens AG, Corporate Technology
Otto-Hahn-Ring 6, 81739 Munich, Germany

***Abstract* - The CERN's accelerator complex and its experiments rely on the proper functioning of a multitude of heterogeneous industrial control systems. Over 600 industrial control systems with more than 40 million sensors, actuators and control objects store more than 100 terabytes of data per year (the volume of generated data is much more). This paper describes a mathematical approach to monitor online a multitude of sensors/actuators and automatically detect signals oscillations. In order to achieve it the presented method combines both expert knowledge and spectrum analysis. Some results, obtained by the application of this analysis to the CERN cryogenics system, are presented showing multiple plant-wide oscillations. Finally the paper briefly describes the deployment of Spark and Hadoop platform into the CERN industrial environment to deal with huge datasets and to spread the computational load of the analysis across multiple hosts.**

Keywords – Expert knowledge system, online fault detection, spectrum analysis, online detection of signal oscillation

I. INTRODUCTION

CERN industrial facilities count about 600 industrial Supervisory Control and Data Acquisition` (SCADA) systems for the supervision and monitoring of its accelerators, detectors and infrastructure machines. The volume of industrial control data stored per year is more than 100 terabytes over 45 million parameters. The generated volume of control data is actually much more since multiple filters (dead-bands both in time and value) are deployed both in the SCADA systems and at the control level (PLCs and FECs) in order to reduce the data flow. The latter contains useful information about the current state of the processes under control, their performance, stability and overall behavior. It is obvious that an extensive analysis of this massive data flow cannot be achieved manually by operators; on the contrary it is necessary to develop and deploy specialized frameworks which are able to provide analytical services and handle the big datasets. The presented algorithm has been developed in line with this vision of industrial analytical services; the automatic detection of signal oscillation has been, indeed, integrated as a continuous monitoring task into the machine operation. Once the analysis can be applied to solve a control problem, it becomes part of the control system itself making possible to raise awareness of operators on specific issues or anomalous conditions from the same supervision application they are accustomed to.

As it has been already pointed out by other studies [1], [2] the detection of anomalies / disturbances in an industrial process represents a key factor in the quality of the overall control system. Oscillations are a common type of plat-wide disturbances whose detection is of importance for all the different industrial control systems at CERN. In order to achieve it the presented method combines both expert knowledge and spectrum analysis. While the experts are knowledgeable about their domain, they are not skilled to perform these data analysis or computing problems tasks. In this context, formalizing expert knowledge means translating the correct operational conditions, known by the machine expert, into simple equations, parameters that can be used as input in the oscillation detection analysis. Then all the analytical process can be scaled out to all similar systems without burdening the users/operators with a vast amount of manual operations to carry out. The amount of data to process is in a range where naive usage of standalone scripts and single node applications reach their limits due to data I/O limitations, memory consumption and CPU load. From a computer science perspective this problem is part of the Big Data field which exploits different technologies for processing huge datasets. This is why the oscillation analysis has been developed as a Spark job to execute against the CERN Hadoop cluster and integrated into our industrial control system.

This paper describes the mathematical approach to detect signals oscillations by combining expert knowledge with

978-1-5090-4253-1/17/$31.00 ©2017 IEEE

spectrum analysis. Then, we will briefly present some oscillatory behaviors detected by analyzing real industrial data; specifically they are related to the valves deployed in the CERN cryogenics control system. At the end of the paper we briefly describes the deployment of Spark and Hadoop platform into the CERN industrial environment.

II. Detection of Signal Oscillations

A. *Oscillations impact and analysis requirements*

Oscillation detection is essential in the evaluation of the performances of any control loops; this is why it has been treated as part of the normal operational analysis. Signal oscillation have multiple impacts in a control system; firstly they can affect the system stability and its safety, which is a necessary feature of any part of the control: e.g. an increase of temperature in the CERN cryogenics system can force the control to quench and lose the bunch of accelerated particles. This means that oscillations can directly affect the physics quality, which is the main goal of production in the CERN industrial environment. Even when oscillations are not the direct cause of a system fault, they can put physical equipment under stress: e.g. a regulation valve opening and closing in an oscillatory behavior; the latter can be directly translated into a higher cost in maintenance (e.g. according to preventive maintenance plan the valves need to be replaced after a specific use expressed in kilometers). At last but not least signal oscillations can increase the communication load and the cost of the storage. In fact multiple filters have been implemented in the various controls systems in order to reduce the amount of exchanged data if the signals are not changing “much” in time and value. Obviously an oscillatory behavior can trespass the thresholds defined in the filters and be the direct cause of increase of communication load and information to store.

The presented method is not based on an accurate model, since the modelling of the CERN large scale control system can be quite expensive to achieve; moreover, unlikely other industrial plants, CERN controls systems are continuously updated and changed (i.e. different LHC machine modes operating with always higher energy); this would mean a continuous work to keep all the models updated. The analysis has to fulfill the following requirements: online detection of oscillations, not detection of transient processes (but oscillation detection during transients), capable of dealing with multi frequency oscillations, and simple in terms of tuning parameters.

III. Oscillation Detection

A. *Background*

The signal oscillation detection analysis [3] represents a clear example of online monitoring activity [4], [5]. Similarly to the operation of the SCADA system this analytical service provides the operators with the possibility to automatically monitor a huge number of sensors/actuators and detect possible faults. The diagnosis of these faults is not part of the analysis itself since an oscillatory behavior can have multiple root-causes like: external disturbances, hardware failures, wrong tuning or structure of the control loop, excessive gain in the feedback, and so on… Therefore the analysis main goal is limited to the detection of the anomalies whose diagnosis will require internal knowledge of the system under analysis and manual operations from the experts.

The analysis cannot rely on control alarms or logs since an oscillatory behavior can take place even when the system is operating in nominal state, so without triggering any control threshold. For instance, in the cryogenics system several abnormal behaviors have been identified by inconsistent sensor readings when the system was running smoothly. Therefore the discovery process can make use of only time series data (such as pressure values or valve position). Due to its generality the analysis can be applied to any control system where signal oscillation detection is of interest. In this specific analysis the attention was focused on the detection of the oscillation of control valves. Under nominal conditions the process values oscillate, causing the valves to open or to close as expected. However, for various reasons these valves may start oscillating with an unexpected frequency or amplitude causing plant-wide perturbation.

B. *The analysis flow*

The analysis flow is represented in the figure below. The developed algorithm consists of a univariate time series analysis which combines both spectral and time-domain analysis. In order to reduce the computational load and be able to run the analysis in online mode a sliding window mechanism is used. Once new signal samples arrive, the Discrete Fourier Transform (DFT) is calculated, on an incremental step, to detect possible peaks in the spectrum. As many references stated [6], [7], [8] this represents a well-known and validated procedure for oscillation detection. Actually the frequency components are compared against a specific threshold. The latter is calculated on the base of expert knowledge according to the procedure described in the following of the document.

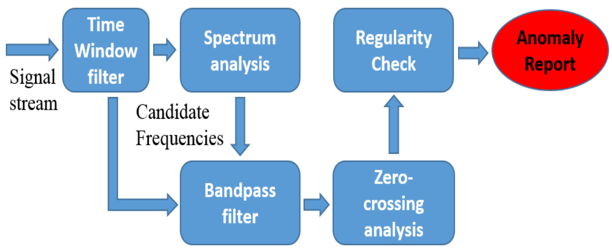

Fig. 1. Analysis flow of the oscillation detection algorithm

As the figure above shows the algorithms has been designed as a multi-steps analysis; this was an expected feature since it reduces the computation time and makes the algorithm fitting an online detection schema. Hence the analysis continues to the next step only for those time windows where the input signals presents frequency peaks.

Obviously the signals sampling rate f_s determines the highest detectable frequency f_{max} of oscillation and the resolution of the frequency spectrum depending on the length of the time window. Even if it is not visible in the schema the Hann-window is applied to the single input signal in order to avoid any side leakage effect in the frequency domain. The discovered peaks in the power spectrum indicate potential oscillation which need to be further investigated in order to neglect false detections (reducing the number of false positives), which for example could instead be due to manual operations or noise. For this purpose these candidate frequencies are then been used as input parameters for a bandpass filter. The main function of the latter is to concentrate the analysis only on a certain frequency interval around the related peak. This frequency interval has been defined according to the zero-crossing rate (as explained later) expected for the relative candidate frequency. This component of the signal in the time domain goes through a zero-crossing analysis [9]. Specifically it is applied to the demeaned version of the filtered signal to check the presence of oscillations. This condition is verified if the number of zero-crossing is higher than a parameterized threshold, the zero-crossing rate. This rate is proportional to the frequency under analysis: lower frequencies will be associated to lower zero-crossing rates. This last checks avoid the false classification of transient processes as oscillations. As a final step a new property is introduced: the regularity in time and amplitude. The regularity of both the oscillation period and amplitude is checked by comparing the standard deviation of the zero-crossing period/amplitude with its mean. For this purpose the method of [10] can be used to indicate if the oscillation is regular both in time and amplitude; it assumes that the standard deviations of the period (std_p) and of the peaks amplitude (std_a) are respectively less than one third of the means of the periods (T_p) and peak values (T_a). For this purpose two simple indexes have been defined as:

$r_p = T_p / (3 * std_p)$,

with $r_p > 1$ *indicating a regular oscillation in time;*

$r_a = T_a / (3 * std_a)$,

with $r_a > 1$ *indicating a regular oscillation in amplitude*

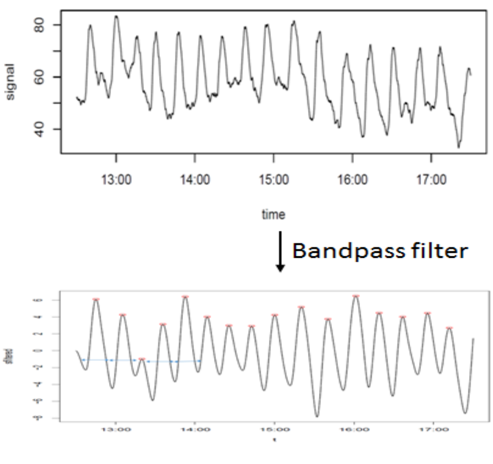

Fig. 2. Zero-crossing analysis and regularity check

In the figure above you can see an input signal (specifically a cryogenic valve) whose frequency component detected as a peak has been used as a parameter of the bandpass filter. After an initial tuning the above algorithm has been able to detect the presence of multiple oscillations with their relative oscillation periods and amplitudes by recognizing regular patterns in the analyzed data. The results will be presented at the end of the paper.

C. *Combining expert knowledge with spectrum analysis*

As described in the previous paragraph a specific threshold is used to detect peaks in the spectrum domain. This threshold is calculated on the base of expert knowledge with a shape of an exponential function. One of the requirement of the algorithm was to provide a simple interface to use and parametrize the analysis. This is why the different experts are asked to provide a list of tuples, L_{ts}; each tuple consists of a combination of amplitude and frequency:

L_{ts} = {*tuple (Amplitude, Frequency)*}

Each tuple represents a specific threshold for an entire system or a specific sensor/actuator. In particular it represents the maximum amplitude of oscillation for a specific frequency; if the signal amplitude overcome the so defined limit then it is considered as a frequency peak. The number of rules per system/signal is not enough to cover the entire spectrum. This is why a regression function with a parametric exponential kernel has been used to define the threshold function which fits best the experts' oscillatory rules against all the frequency spectrum. In the figure below you can see the calculated threshold (red line) which is used to detect frequency peaks by analyzing some valves deployed in the CERN cryogenics system.

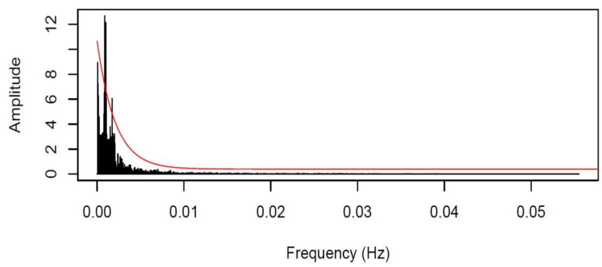

Fig. 3. Definition of the frequency threshold based on expert knowledge

It must also pointed that the threshold function can be easily updated by the introduction of new experts rules or tuples; then a regression analysis will be re-executed to estimate the new threshold function which matches the new list of tuples values (L_{ts}). This threshold definition has been chosen because it can be easily validated by the experts and not hidden behind a mathematical model of analysis; on the contrary it is actually defined in the experts' domain.

It is worth to mention that the attempts to replace the experts' knowledge with some machine learning algorithms were unsuccessful; the main reason of that is the lack of supervised data and the list of tuples was easier to provide by the experts. Moreover the unsupervised algorithms failed due to the systematic behavior of the control system; under specific conditions some valves were always oscillating, so the model was learning it as a nominal behavior under those conditions. The translation of experts' knowledge into a list of tuples (L_{ts}) has demonstrated to be a simple but at the same time effective way to describe the system model operating in nominal conditions.

IV. Parallelization of the Control Analysis

As already mentioned in the previous chapters the huge amount of data produced by the CERN control systems makes ineffective any naive attempt of using standalone scripts or single node applications; they will be hampered by the limited amount of resources, like I/O bandwidth, memory consumption and CPU load. It is necessary to run the developed algorithm against a large dataset of control signals, sensors, and actuators. Consequently a cluster-based computing approach would be able to handle the enormous computational resources needed to run the analysis. This is why the oscillation analysis has been developed as a Spark job to execute against the CERN Hadoop cluster and integrated into our industrial control system. This solution has been designed to parallelize the execution of such algorithm, showing the positive benefits of scaling the analysis across multiple nodes. Moreover the lightweight portability of the spark jobs solves any deployment issues linked to differences in execution environments.

A. *Spark job implementation*

The presented algorithm represents a univariate analysis; therefore there were no constraints to parallelize the execution of multiple signals in different nodes. Moreover a time window approach has been adopted to avoid any continuous reload of the data; on the contrary only the last values, depending of the time window length, were kept in all the nodes memory for the analysis.

Another aspect to mention is the reduction of data shuffling among the nodes. The implementation through the Spark API has been done without the use of any wide operation (that requires the data from two or more different nodes). To achieve this the algorithm makes sure that each node keeps in memory the single signal time window without requiring any shuffling from different nodes. This optimization improved the speed of the analysis by 2x factor.

Spark [11] support both Python and R development languages for data scientists. Unfortunately the Cloudera distribution currently installed at CERN does not support R. This is why Python was chosen, but also because due to the large number of modules or packages that are readily available for signal processing. Specifically Numpy [12] and SciPy [13] have been combined with Spark Python API to deal with multidimensional array, frequency domain functions (e.g. DFT) and algebra operations.

V. Results

This last part of the document shows the results of the analysis applied to the CERN cryogenics system. First a general description of this peculiar control system will be given; then some results and plots showing oscillatory behaviors detection will be proposed.

A. *CERN cryogenics system*

Cryogenics is the branch of physics that studies the production of very low temperatures and their effects. The CERN Large Hadron Collider (LHC) is the largest cryogenic system in the world. A current of 11,850 amps in the magnet coils is needed to produce a magnetic of 8T in order to keep particle beams on course around the LHC's 27-kilometre ring. The LHC's superconducting magnets are therefore maintained at 1.9 K by a closed liquid-helium circuit in order to exploit the superconductivity phenomena (electricity conduction with no resistance) and avoid overheating in the coils. The entire cooling process consists of three different stages and takes almost one month to complete [14]. Any oscillatory behavior can affect the system stability and its safety; the increase of temperature, if not properly controlled, could result in an overheating of the coils forcing the system to quench losing the bunch of accelerated particles. As previously mentioned oscillations put physical equipment, like valves, under stress with the following higher cost in maintenance.

B. *Valve oscillation detection in cryogenics system*

Currently more than 5000 signals from different CERN control systems are being continuously monitored; actually a Spark job is running for oscillation detection. Specifically the results presented in this paper represents different oscillations of the valves deployed in the CERN cryogenics system; nevertheless the same analysis has been used to detect oscillations within the CERN cooling and ventilation plants. Therefore the input signals express the movement (in percentage of opening/closing) of the single valves. The figure below shows a clear oscillatory behavior and the analysis process to detect it: first the spectrum analysis comparing each frequency amplitude against the experts' knowledge - based threshold; then the zero-crossing analysis and the regularity check.

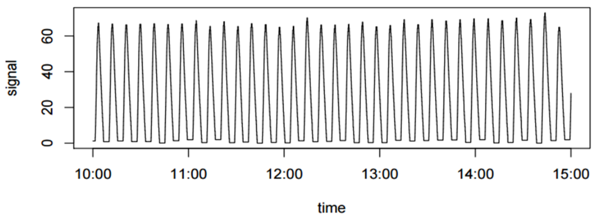

Fig. 4. Movement of the cryogenic valve in time

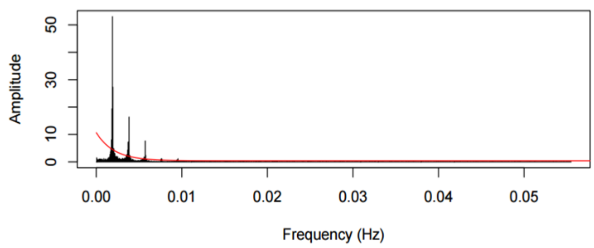

Fig. 5. Spectrum of the input signal against calculated threshold

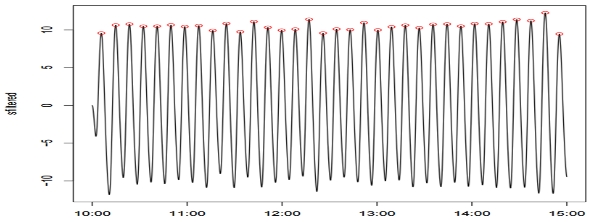

Fig. 6. Zero-crossing analysis and regularity check

As previously stated the analysis stops at the detection of the anomalies; an extra effort is necessary from the operator or system expert to understand the root cause of the anomalous behavior.

The next presented result shows the ability of the algorithm to detect signal oscillations in the whole spectrum range. Unlike vibration analysis it was a requirement to detect any oscillation without focusing on a specific range of frequency. The figure below shows indeed an oscillation in high frequency range and the analytical process to detect it.

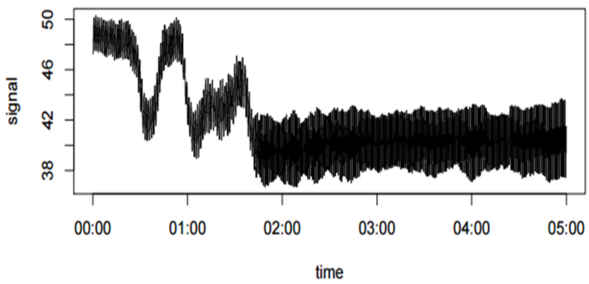

Fig. 7. Movement of the cryogenic valve in time

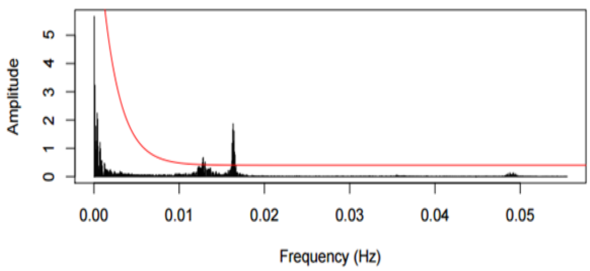

Fig. 8. Spectrum of the input signal against calculated threshold

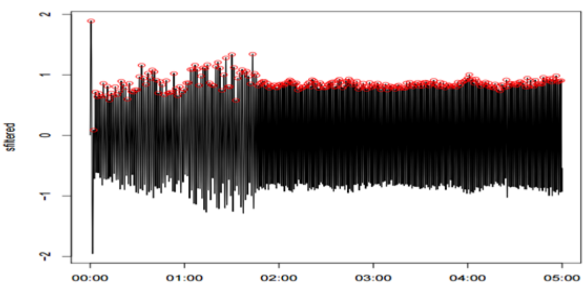

Fig. 9. Zero-crossing analysis and regularity check

Conclusions and Future Work

The work presented in this paper describes a procedure for online detection of signal oscillations. Specifically it combines spectrum analysis with expert knowledge. The easiness of injecting experts' knowledge and formalizing it into simple rules represents one of the major benefits. The analytical method has been proven against real industrial data generated by different CERN control systems. The analysis has been fully automated and it is used to monitor the oscillatory behavior of more than 5000 control sensors and actuators. Once an anomalous oscillation has been detected the operator is provided with: a chart of the signal in the time window when the anomaly was detected, the amplitude of oscillation and its period. This information has been useful to diagnose and correct the system behavior (most of the time by tuning the control loop parameters).

Future work will include reinforcement learning by injecting the analysis experts' feedback on both true and false positives/negatives. In this way the method will automatically integrate experts' knowledge and will better adapt to the expected control process behavior.

Acknowledgment

The presented work has been achieved within the openlab collaboration between CERN and Siemens. A particular thanks to E. Blanco Vinuela, F. Varela Rodriguez, P. Gayet, A. Rijllart, and A. Voitier for their helpful contribution.

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